

THE ROAD TOWARDS PRECISION MEASUREMENTS OF H_0 WITH BRIGHT SIRENS IN THE EINSTEIN TELESCOPE ERA

CHIARA DE LEO 

Physics Department, Sapienza University of Rome, P.le A. Moro 5, 00185 Roma, Italy
INAF - Osservatorio Astronomico di Roma, via Frascati 33, 00078 Monte Porzio Catone, Italy
Istituto Nazionale di Fisica Nucleare (INFN), Sezione di Roma, P.le A. Moro 5, I-00185, Roma, Italy and
Physics Department, Tor Vergata University of Rome, Via della Ricerca Scientifica 1, Roma, 00133, Italy

ELSA M. TEIXEIRA  AND VIVIAN POULIN 

Laboratoire Univers & Particules de Montpellier,
CNRS & Université de Montpellier (UMR-5299), 34095 Montpellier, France
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Abstract

Gravitational-wave standard sirens provide an independent probe of cosmic expansion since their luminosity distances are inferred directly from the gravitational-wave signal and, for bright sirens (BSs), the source redshifts are obtained through the identification of electromagnetic counterparts. In this work, we forecast the constraining power of future bright siren catalogues on the Hubble constant using simulated binary-neutron-star mergers detected by the Einstein Telescope with associated electromagnetic counterparts. We construct mock catalogues with different numbers of events and redshift distributions, with GW170817 as a reference BS, and analyse the resulting constraints within a flat Λ CDM cosmology. We find that, when bright sirens are used as a standalone probe, most of the information on H_0 is provided by low-redshift events, with the improvement in precision saturating for sources above $z \sim 1$; beyond that redshift, the constraining power of BSs is increasingly limited by the degeneracy with Ω_m , which prevents further gains in precision on H_0 . In this case, approximately 90 low-redshift BSs are required to reach $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while about 45 are sufficient for $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$. When external BAO information is included to reduce the $H_0 - \Omega_m$ degeneracy, intermediate-redshift sirens become more informative and the required number of events decreases substantially, to roughly 20 and 15 BSs for $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$, respectively. These results highlight the importance of both electromagnetic counterpart identification and complementary background probes in making BSs a competitive, distance-ladder-independent test of the Hubble tension.

Subject headings: Gravitational waves, Bright sirens, Hubble tension

1. INTRODUCTION

The standard cosmological model, the Λ CDM model, posits the existence of a dark sector needed to explain current observations. Here, Λ is the cosmological constant, a constant dark energy source to explain the late-time phase of accelerated expansion. On the other hand, cold dark matter (CDM) is introduced to provide a mechanism for the formation and dynamics of structures in the Universe. The current expansion rate of the Universe is parametrised by the Hubble constant $H_0 = H(z=0)$, which is the normalisation constant of the background expansion rate:

$$H(z) = H_0 \sqrt{\Omega_{r,0} a^{-4} + \Omega_{m,0} a^{-3} + \Omega_\Lambda}, \quad (1)$$

in a spatially flat FLRW background, where a is the scale factor of the Universe and $\Omega_{r,0}$, $\Omega_{m,0}$, and Ω_Λ are the present fractional densities of radiation, matter (CDM and baryons), and the cosmological constant. A direct probe of the expansion history of the Universe is the luminosity distance, d_L , which for a source at redshift z can be estimated from

$$d_L(z) = c(1+z) \int_0^z \frac{dz'}{H(z')}. \quad (2)$$

The Hubble constant sets the absolute scale of cosmological distances. Current determinations of H_0 show a significant discrepancy between values obtained from the late-Universe distance ladder and those inferred from early-

Universe observations within the Λ CDM model. In particular, the SH_0 ES distance-ladder analysis, based on Cepheid-calibrated Type Ia supernovae, gives $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Riess et al. 2022). On the other hand, observations of the cosmic microwave background (CMB) from the *Planck* satellite, when interpreted assuming Λ CDM, give $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Aghanim et al. 2020). This mismatch is commonly referred to as the Hubble tension and represents one of the most pressing open problems in modern cosmology.

Several explanations have been proposed, depending on whether the tension is interpreted as the result of unaccounted observational systematics, as a hint of limitations in the assumptions made under Λ CDM or as a possible indication of physics beyond the standard model. On the observational side, the discrepancy could be related to residual systematics in the local distance ladder, for example in the calibration of Cepheid variables, in the standardisation of Type Ia supernovae, or in the connection between the different rungs of the ladder (Freedman 2021; Efstathiou 2021). In fact, the recent H_0 Distance Network consensus finds that the local determination is robust across first-rank distance indicators, with a covariance-weighted baseline value $H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$; this makes a simple systematic error associated with a single calibrator, distance indicator, or ladder rung increasingly difficult to isolate as the sole origin of the tension (Casertano et al. 2026). On the theoretical side, if the tension is not due to systematics, it may suggest that some assumptions of the standard cosmological model need to be modified. Proposed solutions can be broadly divided into early-time mechanisms, which affect the pre-recombination Universe and reduce the sound horizon, and late-time mechanisms, which modify the expansion history at low redshift (Verde et al. 2019; Di Valentino et al. 2021; Poulin et al. 2023). In this context, independent probes of the distance-redshift relation are essential to test whether the discrepancy is due to the calibration of electromagnetic distance indicators, to the assumptions of Λ CDM, or to a combination of both.

Bright standard sirens (BSs) provide such an independent route, since they measure luminosity distances directly from the gravitational-wave (GW) signal and obtain redshift information from an electromagnetic (EM) counterpart. For compact binary mergers, the observed waveform encodes the luminosity distance in terms of the amplitude and phase evolution. If an EM counterpart for the event is identified, the redshift of the host galaxy can also be retrieved, producing an independent measurement of the luminosity distance. The first measured BS was from the GW170817 event, a binary neutron star (BNS) merger detected in gravitational waves and associated with the EM counterpart GRB 170817A and the host galaxy NGC 4993 (Abbott et al. 2017; Hjorth et al. 2017; Cowperthwaite et al. 2017). For a single event, the resulting constraint on H_0 is much broader than current electromagnetic (e.g. from supernovae) and CMB determinations, but GW170817 demonstrated the potential of bright siren cosmology (Abbott et al. 2017). Future third-generation detectors, such as the Einstein Telescope (Abac et al. 2026), are expected to detect many more events over a broad redshift range, and a subset of these events may have detectable electromagnetic counterparts, allowing them to be used as BSs.

The potential of gravitational-wave standard sirens to measure H_0 has been extensively investigated in the literature. Early forecasts focused primarily on nearby binary neutron star mergers with identified electromagnetic counterparts, showing that samples of order 50–100 bright sirens observed with the current Ligo–Virgo–Kagra (LVK) detector network could distinguish between the local and CMB-inferred values of H_0 , or reach percent-level precision (Chen et al. 2018; Feeney et al. 2019). Subsequent studies extended these analyses to third-generation observatories such as the Einstein Telescope (ET), considering larger catalogues spanning a broader range of redshifts (de Souza et al. 2022). The complementarity between standard sirens and electromagnetic probes, including BAO, has also been explored as a way of breaking degeneracies in parameters describing the background expansion, such as H_0 and Ω_m (Chang et al. 2019; Zhang et al. 2019) and to constrain extended models (Teixeira et al. 2023; Giarè et al. 2025; De Leo et al. 2026; Graziotti et al. 2026).

The importance of the redshift distribution for BSs has also been the focus of previous work. At sufficiently low redshift, the luminosity distance in Eq. 2 is dominated by the leading-order relation $d_L(z) \approx cz/H_0$, so nearby events provide a comparatively direct estimate of H_0 . At higher redshift, the distance-redshift relation becomes increasingly sensitive to the matter density and to the assumed background history. In the context of third-generation GWs, most of the information on H_0 has been found to be supplied by local events and that sources above $z \sim 1$ provide only limited additional improvement, as shown for dark sirens by Muttoni et al. (2023). These results suggest that the constraining power of standard sirens does not depend solely on the size of the catalogue, but also on its redshift distribution, distance uncertainties, EM counterpart selection function and the availability of complementary cosmological information.

Building on these studies, our aim is not to establish the general low-redshift sensitivity of standard siren measurements, but to quantify it for a population of ET binary neutron star detections with identified EM counterparts. We investigate how different portions of the resulting BS catalogue contribute to the marginalised constraint on H_0 and how this redshift dependence changes when the sirens are combined with mock future BAO measurements from SKA Observatory (SKAO). We then estimate the number and redshift distribution of identified bright sirens required for the resulting measurement of H_0 to reach a precision of $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$, respectively.

Our main result is presented in Fig. 1, where we show the posterior obtained from the only BS event detected to date, GW170817, compared with one of the mock BS catalogues analysed alone, represented by the dashed orange curve, and the corresponding posterior obtained after including external BAO information, represented by the dashed green curve. We find that about 20 BS events detected by ET in combination with a constraint on Ω_m from BAO data can reach $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

This paper is structured as follows. We first introduce cosmology with BSs in Section 2. We then describe the observables and datasets used in the analysis in Section 3. In Section 4, we present the methodology adopted to obtain

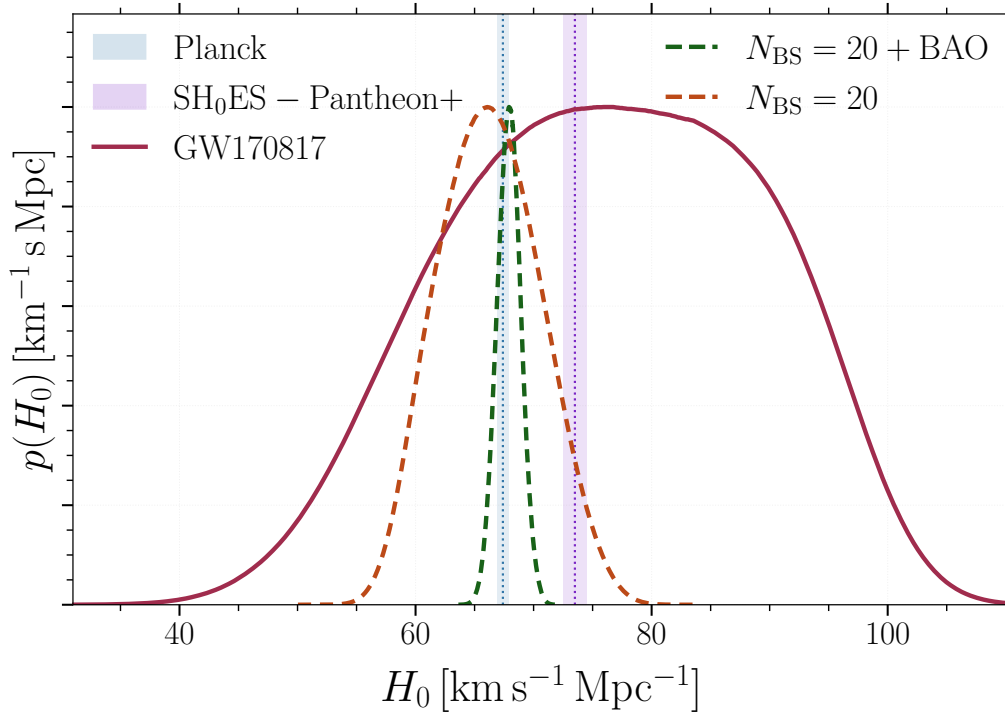


FIG. 1.— Posterior probability distribution of H_0 inferred from GW170817, compared with the SH_0ES -calibrated Pantheon+ H_0 (purple) measurement, the value constrained from *Planck* under a Λ CDM cosmology (blue) and the posterior obtained using 20 mock bright-siren events alone (dashed orange) and analysed with external BAO information (dashed green). The mocks are generated following the prescriptions described in Sections 3.2 and 3.3.

the results, which are discussed in Section 5. Finally, we draw our conclusions in Section 6.

2. BRIGHT SIRENS

Gravitational waves can be described as small perturbations $h_{\mu\nu}$ of the space-time metric. In a flat FLRW expanding Universe, tensor perturbations propagate according to

$$h''_{+,\times}(\tau, k) + 2\mathcal{H}h'_{+,\times}(\tau, k) + k^2 h_{+,\times}(\tau, k) = 0, \quad (3)$$

where primes denote derivatives with respect to conformal time τ , k is the comoving wavenumber, $\mathcal{H} = a'/a$ is the conformal Hubble parameter, and h_+ and $h_\times$ are the two GW polarizations. Here, the friction term describes the damping of the GW amplitude due to the cosmic expansion.

A variety of astrophysical and cosmological mechanisms are expected to generate GWs, including compact binary coalescences (CBCs), core-collapse supernovae, rotating non-axisymmetric neutron stars, stochastic backgrounds of astrophysical or primordial origin, and possible signals from the early Universe. To date, however, all confirmed direct GW detections have been associated with CBCs involving binary black holes, binary neutron stars, and neutron-star-black-hole systems (Abbott et al. 2016, 2017; Abac et al. 2025, 2026). These sources are particularly relevant for cosmology because their waveform provides a direct and independent measurement of the luminosity distance. Unlike EM probes, which rely on the calibration of standard candles or standard rulers, CBCs are self-calibrating sources: the distance information is encoded in the amplitude and phase evolution of the signal. For this reason, they are commonly referred to as *standard sirens* (Schutz 1986; Holz & Hughes 2005).

For a compact binary inspiral, at leading order and in the frequency domain, the strain amplitude scales as

$$\tilde{h}(f_d) \propto \frac{\mathcal{M}_d^{5/6}}{d_L} f_d^{-7/6} e^{i\Psi(f_d, \mathcal{M}_d)}, \quad (4)$$

where f_d is the detector-frame frequency, Ψ is the GW phase, and $\mathcal{M}_d$ is the detector-frame chirp mass. The detector-frame quantities are related to the source-frame ones by

$$f_d = \frac{f_s}{1+z}, \quad \mathcal{M}_d = (1+z)\mathcal{M}_s. \quad (5)$$

Therefore, the GW signal directly constrains the luminosity distance, d_L , and the redshifted chirp mass, $\mathcal{M}_d$. However, because the source redshift cannot be determined from the GW signal alone, an independent redshift measurement is essential for cosmological applications (Maggiore 2018; Palmese & Mastroianni 2025; Pierra & Mastroianni 2025).

When an electromagnetic counterpart, such as a short gamma-ray burst or a kilonova, is identified, the host galaxy can be associated with the GW event and its redshift can be measured directly. These events are known as *bright sirens*. They differ from dark sirens, for which no EM counterpart is observed and the redshift must instead be inferred statistically.

In this work, we focus on bright sirens. They represent the cleanest way of using GWs for cosmology, since each event provides both a GW luminosity distance and an EM redshift. The cosmological information is then extracted by comparing the measured d_L - z relation with the theoretical prediction, according to Eq. 2. At low redshift, this relation is mainly sensitive to the Hubble constant H_0 . For this reason, BSs are particularly relevant in the context of the Hubble tension, since they offer an independent measurement of H_0 that does not rely either on the cosmic microwave background or on the local distance ladder.

3. DATASET

3.1. Current bright-siren data

As discussed in Sections 1 and 2, bright sirens provide an independent measurement of the cosmic expansion and therefore represent a valuable probe of the Hubble tension.

To date, GW170817 is the only gravitational-wave event with an identified electromagnetic counterpart that has provided a bright-siren measurement of H_0 (Abbott et al. 2017). It was the first binary neutron-star inspiral detected by Advanced LIGO and Advanced Virgo, with a combined signal-to-noise ratio of 32.4 and a sky localisation of 28 deg^2 at 90% probability. The GW signal provided a direct measurement of the luminosity distance, $d_L = 40^{+8}_{-14} \text{ Mpc}$, while the subsequent identification of electromagnetic counterparts, including GRB 170817A and the host galaxy NGC 4993, supplied the redshift information, $z \simeq 0.0098$ (Hjorth et al. 2017).

In this work, we analyse GW170817 using the BS likelihood implemented in the CANDI¹ code (De Leo et al. 2025), which combines the measured luminosity distance and source redshift to derive posterior constraints on the cosmological parameters. Although GW170817 established the potential of BSs as independent cosmological probes, its constraint on H_0 remains substantially broader than those obtained from electromagnetic observations. As shown in Fig. 1, the current BS measurement is therefore not yet competitive with either the distance-ladder determination or the CMB-inferred constraint.

This naturally raises the question of how many additional BSs would be required for a GW-based measurement of H_0 to become competitive with current electromagnetic constraints from SH_0 ES-Pantheon+ and *Planck*. Reaching this level of precision would make BSs a powerful independent tool for investigating whether the observed discrepancy arises from astrophysical systematics or assumptions about the cosmological model.

To address this question, in the following section we construct mock catalogues of future BS events and analyse them both independently and in combination with external mock BAO information.

3.2. The bright sirens catalogue

The main goal of this work is to assess how many BS events are required to constrain the Hubble constant H_0 with a precision competitive with current electromagnetic constraints, in particular the SH_0 ES-calibrated Pantheon+ (Riess et al. 2022) and *Planck* (Aghanim et al. 2020). To this end, we build a mock catalogue of BNS mergers with associated electromagnetic counterparts. The full pipeline used to generate the catalogue is publicly available in the code CANDI. Our simulations are based on the expected performance of the Einstein Telescope (Abac et al. 2026), a forthcoming third-generation ground-based GW detector. We follow the specifications described in Branchesi et al. (2023), assuming a triangular configuration with 10 km arms located in Sardinia, among the possible configurations. We note that alternative design choices, such as adopting two L-shaped interferometers instead of a triangular layout, can change the detector sensitivity and the expected event rate (Branchesi et al. 2023). These differences may therefore affect the number and redshift distribution of the detected BSs and, consequently, the quantitative forecasts presented in this work.

We generate $N = 20000$ injected GW signals from BNS mergers in the redshift range $0.1 < z < 3.5$. This interval is chosen to reproduce the BNS detections expected for the Einstein Telescope (Branchesi et al. 2023). We note, however, that the lower cut at $z = 0.1$ excludes the very nearby population. Therefore, GW170817 will always lie below the redshift range of our simulated catalogues, thereby providing the lower-redshift reference point throughout our analysis. This is a limitation of the adopted ET-oriented mock construction, rather than an indication that bright sirens cannot occur at lower redshift.

The source population is distributed according to a redshift-dependent merger rate, following Cutler & Holz (2009); Hogg et al. (2020),

$$R(z) = \begin{cases} 1 + 2z, & z \leq 1, \\ \frac{3}{4}(5 - z), & 1 < z < 5, \\ 0, & z \geq 5. \end{cases} \quad (6)$$

The corresponding redshift distribution is given by

$$P(z) \propto \frac{R(z) 4\pi d_C^2(z)}{H(z)(1+z)}, \quad (7)$$

¹ <https://github.com/chiaradeleo1/CANDI>

where $d_C(z)$ is the comoving distance and the factor $(1+z)^{-1}$ accounts for the conversion between source-frame and observer-frame time. The inclination angle ι is drawn assuming $\cos \iota$ uniformly distributed in the interval $[-1, 1]$. The events are simulated using a modified version of the public code **darksirens**² (Martinelli et al. 2022). The code implements the merger rate in Eq. (6) and assumes a monochromatic neutron-star mass of $1.4 M_\odot$.

For each event, the GW luminosity distance d_L^{GW} is computed using the theory module of **CANDI**. As fiducial cosmology, we adopt the best-fit values obtained from *Planck*+BAO (Aghanim et al. 2020) in Λ CDM. Specifically, we use

$$H_0 = 67.66 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_m = 0.3111. \quad (8)$$

The uncertainties on the GW luminosity distance and the detectability of the events are evaluated using the public code **GWFish**³ (Dupletsa et al. 2023). We apply a signal-to-noise-ratio threshold and consider an event as detected if $\text{SNR} > \text{SNR}_{\text{th}} = 20$, which corresponds to the most conservative detectability threshold considered for ET in Branchesi et al. (2023).

For the electromagnetic counterparts, we focus on the afterglow emission associated with short gamma-ray bursts (GRBs), while we do not include kilonova emission (Metzger 2019). The GRB emission is produced by a highly collimated relativistic jet, which generates a prompt gamma-ray signal followed by a broadband afterglow extending from X-rays to radio wavelengths. The afterglow can remain observable over long timescales, up to years after the merger. Kilonova emission, instead, arises from neutron-rich ejecta and is typically more isotropic, peaking in the optical or infrared bands depending on the ejecta composition.

To model the detectability of the EM counterpart, we follow the approach presented in De Leo et al. (2025). In that work, the Vera Rubin Observatory/LSST sensitivity (Bianco et al. 2022) was adopted as a reference for EM detectability, and the afterglow emission was modelled using the code **afterglowpy**⁴ (Ryan et al. 2020). It was shown (see Figure 3 of De Leo et al. (2025)) that, in terms of the final number of detectable counterparts, this procedure is approximately equivalent to applying a cut on the binary inclination angle. Following this result, we select BSs by imposing $\iota < \iota_{\text{cut}} = 18^\circ$.

To perform the analysis presented in Section 5, and in particular to study how the constraint on H_0 depends on both the number of BSs and their redshift distribution, we construct several sub-catalogues from the master catalogue, each containing a different number of events. The detailed properties of these sub-catalogues are discussed in the corresponding parts of Section 5. All sub-catalogues include GW170817, which represents the $N = 1$ case. Larger catalogues are then built by adding $N - 1$ events randomly selected from the master catalogue. The catalogues are constructed progressively, so that samples with larger N contain the events included in the smaller samples plus additional randomly selected BSs. This procedure allows us to track how the uncertainty on H_0 evolves as the number of BS events increases. We emphasise that by including the observed GW170817 we are combining simulated measurements using the forecast ET sensitivity with a real LIGO-Virgo data point. This approach is statistically consistent since all the events are treated separately (diagonal covariance matrix) and we have considered a Gaussian distance uncertainty for GW170817. Nevertheless, this means that the smallest catalogues can be affected by the particular noise realisation, while the larger catalogues are progressively dominated by the simulated ET population.

3.3. The baryon acoustic oscillations mock catalogue

Bright sirens alone cannot fully disentangle the effects of H_0 and Ω_m in the background cosmology. This can be understood from the fact that BSs constrain the expansion history through the luminosity distance, which depends on the Hubble function. In a flat Λ CDM background, neglecting the radiation contribution, Eq. 1 can be rewritten as

$$H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + (1 - \Omega_m)}. \quad (9)$$

Since $H(z)$ depends on both H_0 and Ω_m , these parameters can be partially degenerate in a BS-only analysis. For this reason, we include external information on Ω_m from baryon acoustic oscillations (BAO).

The complementarity between standard sirens and BAO in reducing the H_0 - Ω_m degeneracy has already been discussed in previous studies (Chang et al. 2019; Zhang et al. 2019). As we show in Section 5, adding BAO information not only tightens the final constraint on H_0 , but also changes which part of the BS catalogue carries useful information on this parameter, making intermediate-redshift events more informative once the degeneracy with Ω_m is reduced.

BAO measurements constrain combinations of cosmological distances rescaled by the sound horizon at the drag epoch, r_d . In a flat Universe, the relevant distance measures are

$$d_M(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}, \quad d_H(z) = \frac{c}{H(z)}, \quad d_V(z) = [z d_M^2(z) d_H(z)]^{1/3}, \quad (10)$$

where $E(z) = H(z)/H_0$. BAO catalogues provide measurements of d_M/r_d , d_H/r_d , and their combination through d_V/r_d .

Such measurements, for example from galaxy and Lyman- α surveys by SDSS (de Mattia et al. 2021) and DESI (Abdul Karim et al. 2025), are routinely used for precision cosmology. In our forecast, however, we use a mock BAO catalogue rather than current measurements, since the bright siren sample considered here corresponds to a future

² <https://gitlab.com/matmartinelli/darksirens>

³ <https://github.com/janosch314/GWFish>

⁴ <https://github.com/geoffryan/afterglowpy>

ET-era dataset. We adopt an SKAO-like forecast as a representative future BAO probe, allowing us to quantify how complementary geometric information can reduce the $H_0 - \Omega_m$ degeneracy without tying the result to a particular existing BAO data release. Following De Leo et al. (2025), we adopt the optimistic SKA2 configuration (Square Kilometre Array Cosmology Science Working Group et al. 2020) and assume the same fiducial cosmology used for the BS catalogue, described in Section 3.2. The mock measurements cover the redshift range $z \in [0.2, 2]$, divided into 18 bins with width $\Delta z = 0.1$.

Since BAO constrain distances only through their ratio with r_d , additional assumptions would be required to extract absolute distances. In this work, we avoid introducing external priors on r_d , such as those from BBN, in order to keep the analysis independent of early-Universe calibration assumptions. We therefore directly simulate measurements of d_M/r_d and d_H/r_d , propagating the expected relative errors from Bull (2016).

4. METHODOLOGY

The goal of this work is to assess the requirements for future BS detections to reach a precision on H_0 competitive with the SH_0 ES-calibrated Pantheon+ (Riess et al. 2022) and *Planck* (Aghanim et al. 2020). In this way, BSs could become an independent cosmological observable, not relying on the distance ladder, and thereby providing new insight into the Hubble tension.

To this end, we reconstruct the posterior distribution of H_0 using the different sub-catalogues described in Section 3. The inference is performed within a Bayesian framework (Trotta 2017), and the comparison between the simulated data and theoretical predictions is carried out using the public Bayesian analysis code Cobaya⁵ (Torrado & Lewis 2021). For this purpose, we construct external likelihood modules able to handle the datasets discussed in Section 3.

The mock BS and BAO datasets are analysed using dedicated likelihood modules implemented within the CANDI framework. Within the Bayesian approach, the posterior probability distribution of the cosmological parameters θ , given a data vector $\mathbf{d}$, is obtained through Bayes' theorem,

$$\mathcal{P}(\theta | \mathbf{d}) = \frac{\mathcal{L}(\mathbf{d} | \theta) \pi(\theta)}{\mathcal{Z}}, \quad (11)$$

where $\mathcal{L}(\mathbf{d} | \theta)$ is the likelihood function, $\pi(\theta)$ denotes the prior distribution of the cosmological parameters, and $\mathcal{Z}$ is the Bayesian evidence. Assuming Gaussian-distributed measurements, each likelihood can be expressed as

$$-2 \ln \mathcal{L} = [\mathbf{d} - \mathbf{t}(\theta)]^T \mathbf{C}^{-1} [\mathbf{d} - \mathbf{t}(\theta)], \quad (12)$$

where $\mathbf{t}(\theta)$ is the theoretical prediction corresponding to the sampled cosmological parameters and $\mathbf{C}$ is the covariance matrix of the data. The data vector used in the likelihood contains the mock bright siren luminosity distances and BAO distance ratios, $\mathbf{d} \equiv \{d_L(z), d_M(z)/r_d, d_H(z)/r_d\}$, constructed as described above. For the bright siren catalogue, we assume that the luminosity distance measurements of distinct events are statistically independent, meaning that $\mathbf{C}$ is taken to be diagonal. This approximation neglects possible correlated systematic uncertainties, such as a common detector-calibration error. For the BAO mock, we neglect correlations between different redshift bins, but include the forecast covariance between the transverse and radial measurements, d_M/r_d and d_H/r_d within each bin.

For the theoretical modelling, we compute the standard cosmological quantities using CAMB⁶ (Lewis et al. 2000; Howlett et al. 2012), interfaced with a theory module that computes the background quantities. The full setup, made publicly available as CANDI⁷, allows Cobaya to perform Markov Chain Monte Carlo (MCMC) sampling and reconstruct the posterior distribution of the cosmological parameters (Lewis & Bridle 2002; Lewis 2013).

In all analyses, H_0 and Ω_m are treated as free cosmological parameters, with uniform priors over the intervals $H_0 \in [20, 100] \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m \in [0.1, 0.9]$. When BAO measurements are included, we do not impose any BBN information and instead sample directly over the sound horizon at the drag epoch, r_d with a uniform prior $r_d \in [100, 200] \text{ Mpc}$.

5. RESULTS

In Section 3.1, we showed that GW170817 provides the first proof that bright sirens can be used as an independent cosmological probe. However, as illustrated in Fig. 1, the current constraint on H_0 remains significantly broader than those obtained from electromagnetic observations, highlighting the need for substantially larger BS catalogues. In this section, we investigate how many BS detections will be required to achieve a precision of approximately $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ on H_0 , comparable to that of current leading cosmological probes. We also explore whether the constraining power of BSs depends solely on the number of detected events or whether other factors, such as the redshift distribution of the sources and the inclusion of complementary cosmological observations, play a significant role in improving the measurement.

5.1. Impact of the redshift of events

We first investigate how the constraint on H_0 depends on the number of BSs and on their redshift distribution. As described in Section 3.2, we construct a master mock catalogue and use it to generate several sub-catalogues with different numbers of events N , always including GW170817.

⁵ <https://github.com/CobayaSampler/cobaya>

⁶ <https://camb.readthedocs.io/>

⁷ <https://github.com/chiaradeleo1/CANDI>

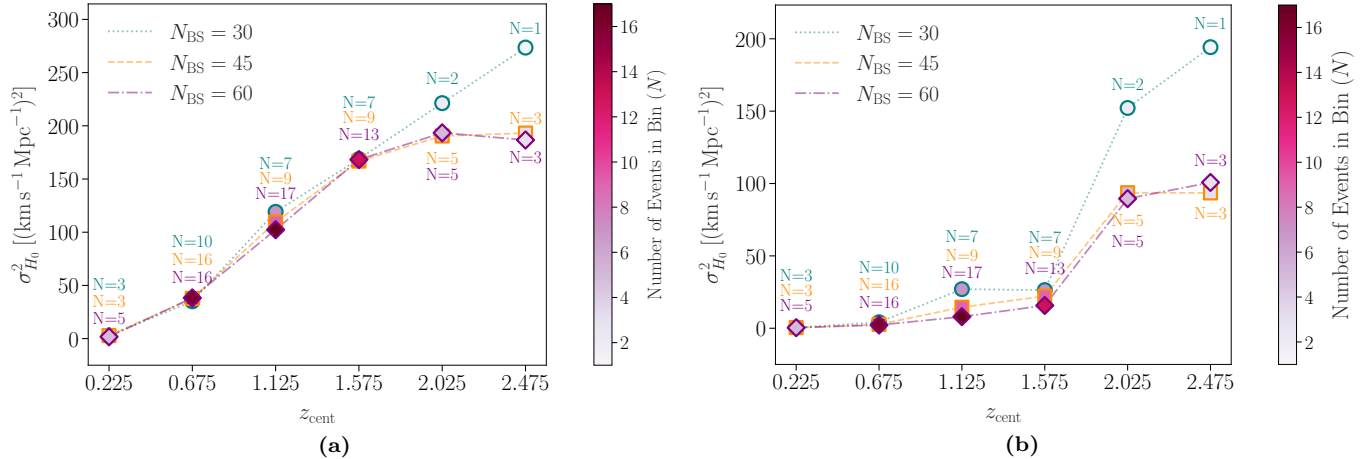


FIG. 2.— Variance of the inferred H_0 posterior as a function of redshift bin for three mock bright siren catalogues containing $N = 30$, $N = 45$, and $N = 60$ events. The events are divided into six equally spaced redshift bins centred at z_{cent} and the labels indicate the number of events in each bin. Panel (a) shows the results obtained when only BS data are included, while panel (b) shows the corresponding results when complementary BAO information is included.

To assess the role of redshift, for each value of N we divide the events into six equally spaced redshift bins between z_{min} and z_{max} . These limits are defined from the largest sub-catalogue considered in this part of the analysis, containing $N = 60$ events. Each bin is then analysed separately by sampling for H_0 and Ω_m only, and computing the variance of the marginalised H_0 posterior, $\sigma_{H_0}^2$. The results are shown in Fig. 2 for three representative catalogues with $N = 30$, $N = 45$, and $N = 60$ events.

The binned analysis reveals a clear dependence of the marginalised H_0 constraint on redshift. As shown in panel (a) of Fig. 2, the lowest redshift bin, centred around $z_{\text{cent}} \simeq 0.23$, consistently yields the smallest variance on the H_0 posterior. The variance increases as we move towards higher redshift for all three catalogue sizes, indicating that events in the lower-redshift part of the catalogue provide a more direct constraint on H_0 . This trend cannot be attributed simply to the number of events in each bin since, for example, the third bin contains 7, 9, and 17 events in the $N = 30, 45$ and 60 catalogues, respectively, but the resulting variances remain of comparable magnitude, suggesting that adding events in this redshift range produces only a limited improvement in the marginalised constraint on H_0 .

A further feature emerging from this analysis is that, at high redshift, the variance tends to approach a plateau. This behaviour is visible for the larger catalogues, while it is less evident for the $N = 30$ case, where the number of events in the last two bins is too small. These results indicate that the most informative events for constraining H_0 are those at $z \lesssim 1$, where the variance is smaller and the degeneracy with Ω_m is less severe.

This behaviour is qualitatively expected: as already discussed in the introduction, the luminosity distance becomes increasingly sensitive to Ω_m at higher redshift, while its statistical uncertainty also grows with distance, both of which degrade the amount of information a given event carries on H_0 . The main result of this analysis is to make this expectation quantitative for our BS population: rather than declining smoothly with redshift, the constraining power on H_0 is concentrated at $z \lesssim 1$, beyond which additional events add comparatively little to the marginalised constraint.

We repeat the analysis cumulatively: instead of independent redshift bins, we include all events below a varying maximum redshift z_{max} and track how the marginalised uncertainty σ_{H_0} evolves as the catalogue is extended outward in redshift. This offers an independent cross-check of the binned result above. The results for the $N = 30$ and $N = 60$ catalogues are shown in Fig. 3. When only BS data are considered, as shown in panel (a), σ_{H_0} decreases rapidly as the lowest-redshift events are progressively included and then largely saturates by $z_{\text{max}} \simeq 1$. This does not imply that higher-redshift bright sirens carry no cosmological information, but rather that their additional contribution to the marginalised constraint on H_0 is much smaller than that already provided by the lower-redshift events.

5.2. Impact of the number of events

Taken together, the binned and cumulative analyses show that the lower redshift portion of the catalogue carries most of the standalone information on H_0 , while at high redshift, luminosity-distance uncertainties are larger and the inference of H_0 becomes more degenerate with Ω_m . For this reason, we restrict the mock catalogues used in the following BS-only forecast to $0.1 < z \leq 1$. This choice is intended to retain only the events that contribute more efficiently to the marginalised H_0 constraint. These catalogues are generated following the same procedure adopted for the master catalogue described in Section 3.2, filtering the events according to the redshift range of interest. Our aim is to determine how many BSs are required, within this most informative redshift range, to achieve uncertainties comparable to those of current H_0 measurements, focusing in particular on $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The resulting constraints on H_0 as a function of the number of events are shown in Fig. 4.

The analysis shows a clear improvement in the precision of H_0 as the number of low-redshift BSs increases. The GW170817-only case yields a very broad posterior, which is insufficient to discriminate between the different reference values of H_0 (Abbott et al. 2017). The uncertainty already begins to decrease noticeably when only a few additional

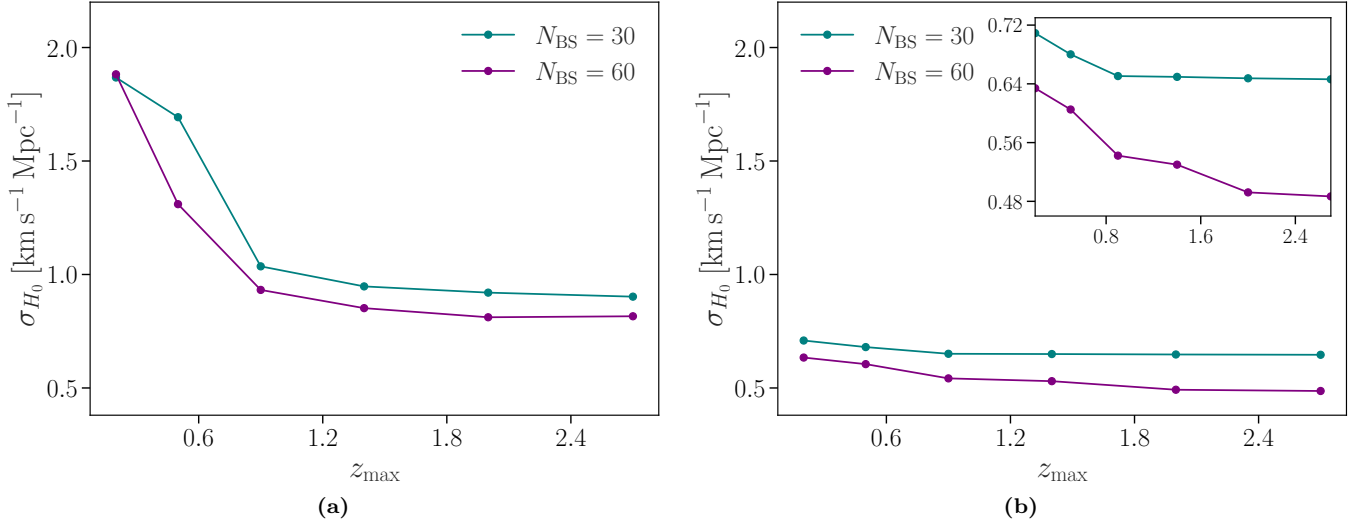


FIG. 3.— Marginalised uncertainty on H_0 as a function of the maximum catalogue redshift threshold z_{max} , for catalogues containing $N = 30$ and $N = 60$ events. Each point includes all events with $z \leq z_{\text{max}}$. Panel (a) shows the results obtained when only BS data are analysed, while panel (b) shows the corresponding results when complementary BAO information is included.

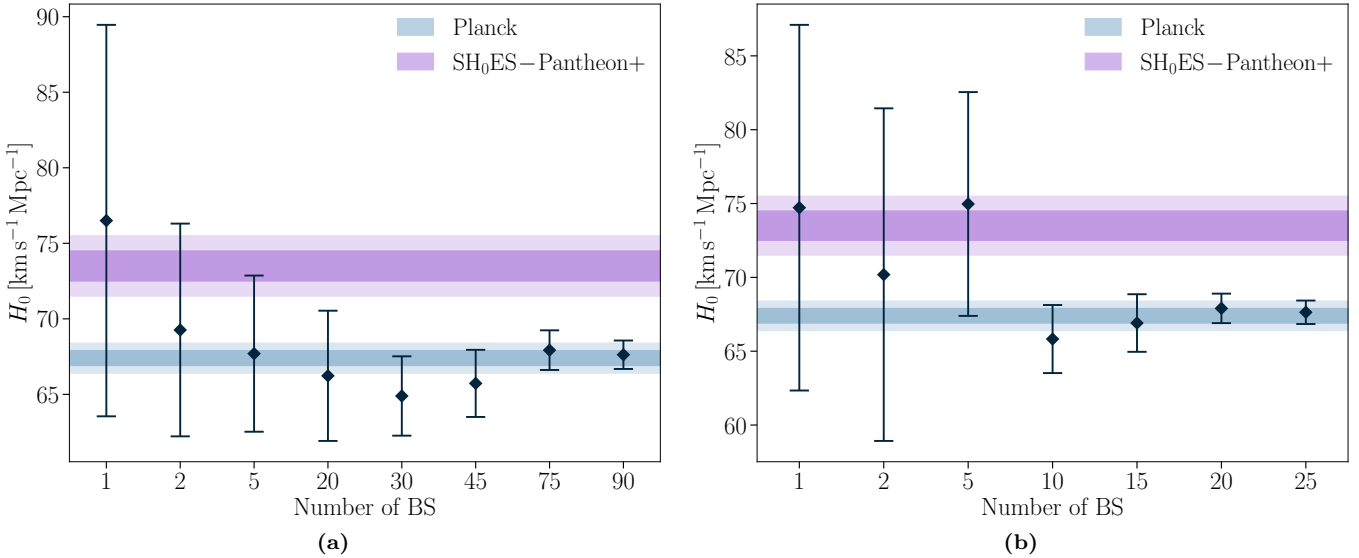


FIG. 4.— Constraints on H_0 as a function of the total number of bright sirens, including GW170817. Panel (a) uses bright siren catalogues restricted to the redshift range $0.1 < z \leq 1.0$, with both H_0 and Ω_m being sampled. Panel (b) combines catalogues extending to $z \leq 3$ with external information from the SKAO BAO mock catalogue, while varying H_0 , Ω_m and r_d . The purple bands show the SH₀ES-Pantheon+ constraint, with the darker and lighter regions corresponding to the 1σ and 2σ intervals, respectively. The blue bands show the corresponding Planck reference intervals.

events are included, although these small catalogues still do not provide a competitive constraint. The improvement becomes progressively more pronounced as the catalogue grows, with the posterior width continuing to shrink across the different values of N_{BS} considered. Note that, because the mock catalogues are generated using the Planck best-fit value of H_0 , the posterior means fluctuate around the fiducial Planck value. The relevant forecast quantity is therefore the width of the posterior, rather than its overlap with either the Planck or the SH₀ES-Pantheon+ value. In the redshift range $0.1 < z \leq 1.0$, we find that ~ 90 and ~ 45 bright sirens are required for the uncertainty to reach $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{Mpc}^{-1}$ and $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{Mpc}^{-1}$, respectively.

Thus, although low-redshift bright sirens provide a clean and independent measurement of H_0 , a catalogue substantially larger than the single event currently available is required for a BS-only analysis to reach the precision of present electromagnetic and CMB-based constraints.

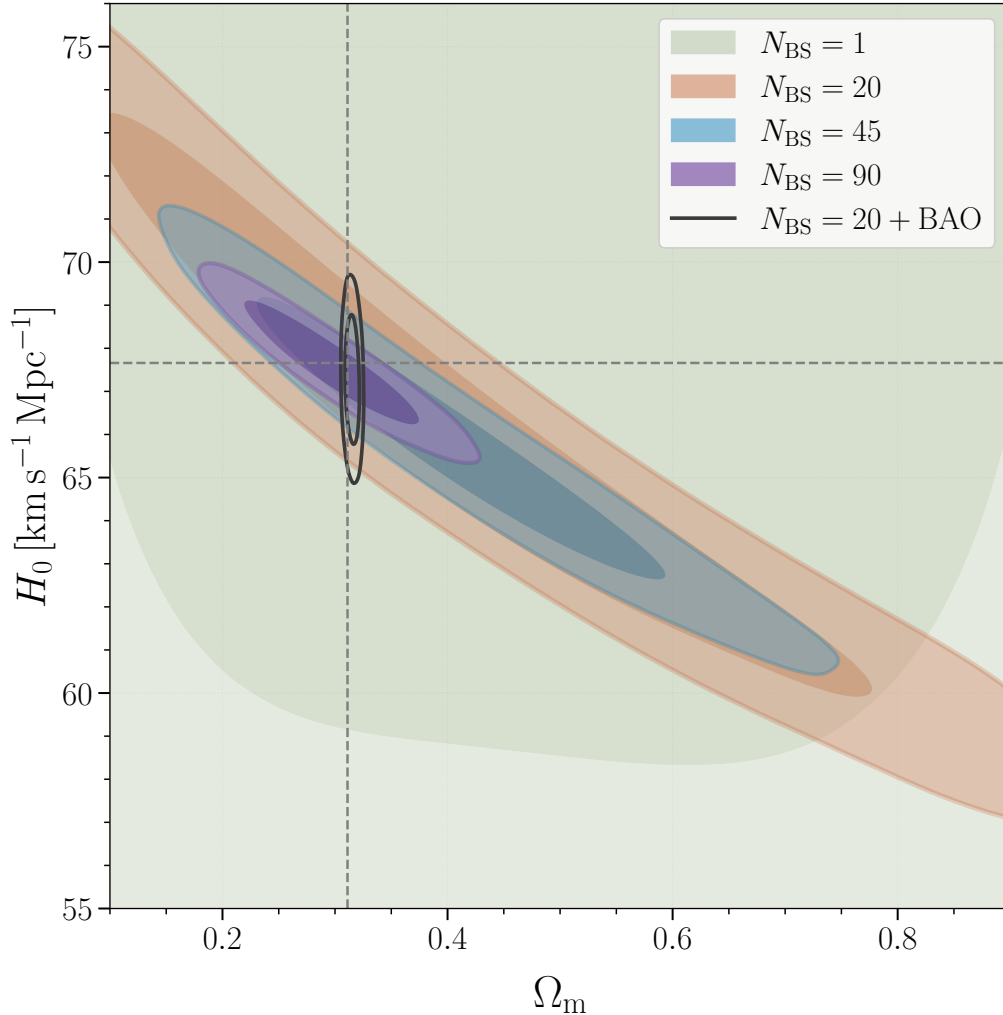


FIG. 5.— Two-dimensional constraints in the H_0 – Ω_m plane obtained from BS catalogues containing $N_{\text{BS}} = 1, 20, 45$, and 90 events, together with the constraint from the $N_{\text{BS}} = 20$ catalogue combined with BAO measurements. The darker and lighter coloured contours refer to the 1σ and 2σ intervals, respectively. The $N_{\text{BS}} = 1$ case corresponds to GW170817, while the remaining catalogues are constructed by progressively adding simulated events to it, as described in Section 3.2.

5.3. Breaking the H_0 – Ω_m degeneracy

The degeneracy between Ω_m and H_0 is illustrated in Fig. 5. In a flat Λ CDM cosmology, the luminosity distance depends both on the overall distance scale set by H_0 and the shape of the expansion history determined by Ω_m . This degeneracy is weak for sufficiently nearby sources but becomes increasingly important as the catalogue is extended to higher redshifts. This is where a constraint on Ω_m from, e.g., BAO data can help: by breaking the degeneracy, it significantly improves the constraining power of high-redshift BSs on H_0 .

We first repeat the binned-redshift analysis after including the SKAO BAO mock. As shown in panel (b) of Fig. 2, complementary BAO information reduces the uncertainty on H_0 at all redshifts, with the largest relative gain in the intermediate and high-redshift bins: once BAO data restrict the background expansion history, the same bright sirens retain substantially more information on H_0 than in the BS-only case. Concretely, the informative redshift range extends to $z \sim 1.5$, compared with $z \lesssim 1$ when BS are analysed alone. The variance still increases for the most distant events, whose distance uncertainties are intrinsically larger, but it remains comparable to the uncertainty found at $z \sim 1$ in the BS-only analysis.

The corresponding cumulative analysis, shown in panel (b) of Fig. 3, confirms this picture. Unlike the BS-only case, σ_{H_0} shows no pronounced saturation at $z_{\text{max}} \sim 1$: it continues to decrease as intermediate-redshift events are added, with a gain that shrinks only gradually, so that sources out to $z_{\text{max}} \sim 2$ still contribute visibly. This behaviour is also reflected in Fig. 5, where the black $N_{\text{BS}} = 20 + \text{BAO}$ contours are substantially narrower than the corresponding BS-only contours, demonstrating that the inclusion of BAO information significantly reduces the H_0 – Ω_m degeneracy. Complementary BAO information therefore does more than tighten the final constraint, it also reshapes the redshift range over which the BS catalogue contributes efficiently to the H_0 constraint.

Motivated by this result, for the BS-BAO analysis we extend the mock catalogue up to $z_{\text{max}} = 3.0$. The resulting

constraints are shown in panel (b) of Fig. 4.

First, we note that the substantial gain in precision between the $N = 5$ and $N = 10$ BS catalogues is partly driven by the specific realisation. Since the sub-catalogues are constructed by randomly adding individual events, the properties of the few sources introduced in this interval—such as a particularly well-localised event or relatively high redshift events—can have an outsized impact when the sample is still small. The overall improvement is nevertheless physically expected, since the H_0 – Ω_m degeneracy can be efficiently reduced once the BS contour becomes sufficiently narrow to combine effectively with the BAO constraint. However, the sharp knee observed at $N = 10$ should not be over-interpreted as a fixed, reproducible number.

We find that approximately $N = 20$ and $N = 15$ events are sufficient to reach a precision of $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$, respectively. The comparison highlights the central result of this work: the inclusion of information from BAO on Ω_m enables a substantial improvement with respect to the BS-only case, where we found that about $N \sim 90$ events are required to reach the same level of precision.

6. CONCLUSIONS

In this work, we investigated the observational requirements for future bright siren catalogues to constrain the Hubble constant with a precision comparable to current electromagnetic and CMB measurements. In particular, we studied how the constraining power depends on the number of detected bright sirens, their redshift distribution, and the inclusion of external cosmological information.

We first analysed the impact of the number and redshift distribution of the events by constructing several sub-catalogues from a common master catalogue, each containing a different number of bright sirens. We divided the events into six redshift bins and evaluated the marginalised variance of the H_0 posterior in each bin.

The binned analysis showed that the variance of H_0 increases significantly at $z \gtrsim 1$. Consequently, adding events at these redshifts produces only a limited improvement in the marginalised constraint on H_0 . This behaviour was confirmed by the cumulative analysis, in which events were progressively included up to increasing values of the maximum redshift. We found that the uncertainty decreases rapidly when low-redshift events are added, but becomes nearly constant once sources beyond $z \simeq 1$ are included. This behaviour reflects two compounding effects: the increasing sensitivity of the luminosity distance to Ω_m at higher redshift, and the growing uncertainty on the distance measurement itself for more distant sources. Since bright sirens alone cannot efficiently break the resulting H_0 – Ω_m degeneracy, high-redshift events carry comparatively little information on H_0 once Ω_m is left free to vary.

Motivated by these findings, we restricted the final bright siren-only catalogue to the range $0.1 < z \leq 1.0$ and constructed sub-catalogues with progressively increasing numbers of events. We found that approximately $N \sim 45$ and $N \sim 90$ bright sirens are required to reach a precision of $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$, respectively.

We then repeated the analysis by including external information from the SKAO-like BAO mock catalogue. The binned and cumulative analyses showed that, once the H_0 – Ω_m degeneracy is reduced, bright sirens at intermediate and higher redshifts provide substantially more information on H_0 : the uncertainty continues to decrease out to $z \sim 2$, rather than saturating at $z \simeq 1$ as in the BS-only case, although the most distant events remain limited by their larger luminosity-distance uncertainties. The inclusion of BAO information substantially reduces the number of bright sirens required to reach the precision of current cosmological probes. We find that approximately $N \sim 15$ and $N \sim 20$ events are sufficient to constrain H_0 with uncertainty $\sigma_{H_0} \sim 2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\sigma_{H_0} \sim 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$, respectively.

Overall, our results show that bright sirens are a promising independent probe of H_0 , in particular in the context of the Hubble tension. However, their constraining power is driven not only by the total number of detected events, but also by their redshift distribution and by degeneracies with the other cosmological parameters. Combining bright sirens with complementary probes such as BAO is therefore particularly valuable: it extends the redshift range over which the catalogue is informative and substantially reduces the number of events needed for a competitive measurement of H_0 .

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